
Unit 1: Foundations and Prerequisites

Cheat Sheet and Review Guide

Built on the Math Confidence Reset Method

First principles before formulas, clarity before memorization

Staples Education

Ordinary Differential Equations Roadmap

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1 Part I: Condensed Cheat Sheet

1.1 The Calculus Toolkit Review

The Core Logic

A derivative measures an instantaneous rate of change, that is, the slope of a curve at a single point. An integral measures accumulation, the running total of a quantity. Differentiation and integration are inverse operations, so each one undoes the other. Master these two ideas and every rule below becomes a shortcut rather than a mystery.

The Standard Form

Power rule: $\frac{d}{dx} x^n = n x^{n-1}$

Chain rule: $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$

Product rule: $\frac{d}{dx} (uv) = u'v + uv'$

Power rule for integrals: $\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$

1.2 The Number e

The Core Logic

The number e is the one base for which the exponential function is exactly its own derivative. This single property is the reason e appears in every model of natural growth, decay, and continuous change.

The Standard Form

Limit definition: $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$

Continuous compounding: $A = P e^{rt}$

Self replicating rate: $\frac{dy}{dt} = y \implies y = C e^t$

1.3 Euler's Formula and Identity

The Core Logic

Euler's formula reveals that a complex exponential is a rotation. The expression $e^{i\theta}$ is the point on the unit circle reached by turning through an angle θ .

1.4 ODE Vocabulary

The Standard Form

- **Order:** The highest derivative that appears.
- **Degree:** The power of that highest derivative.
- **Linearity:** y and its derivatives only appear to the first power.

2 Part II: Review Guide

1. Differentiate $f(x) = 7x^4 - 3x^2 + 5$.
2. Differentiate $g(x) = (2x^3 + 1)^5$ using the chain rule.
3. Differentiate $h(x) = x^3e^x$ using the product rule.
4. Evaluate $\int (6x^2 - 4x + 1) dx$, including C .
5. State the limit definition of e and explain why $dy/dt = y$ is solved by an exponential.
6. Solve $\frac{dy}{dt} = ky$ with $y(0) = y_0$.
7. State Euler's formula, then evaluate $e^{i\pi}$.

8. Multiply $2e^{i\pi/6}$ by $3e^{i\pi/3}$ in rectangular form.
9. Classify $y'' + 3y' + 2y = e^x$ and $(y')^2 + y = 0$ by order, degree, and linearity.
10. Eliminate constants from $y = A \cos x + B \sin x$.

3 Part III: Answer Key and Worked Solutions

Solution 1

$$f'(x) = 28x^3 - 6x$$

Solution 2

$$g'(x) = 30x^2(2x^3 + 1)^4$$

Solution 3

$$h'(x) = x^2 e^x (3 + x)$$

Solution 4

$$2x^3 - 2x^2 + x + C$$

Solution 5

$$e = \lim_{n \rightarrow \infty} (1 + 1/n)^n. \quad e^t \text{ is its own derivative.}$$

Solution 6

$$y = y_0 e^{kt}$$

Solution 7

$$e^{i\pi} = -1$$

Solution 8

$$zw = 6i$$

Solution 9

1: Order 2, Degree 1, Linear. 2: Order 1, Degree 2, Nonlinear.

Solution 10

$$y'' + y = 0$$